

Facilitator Guide

TAKE NOTE

This Manual should be used in conjunction with the Learner Guide and the Learner Workbook

Unit Standard Title: Investigate life and work related problems using data and probabilities
Unit Standard No: 9012
Unit Standard Credits: 5
NQF Level: 3

Time allocation

Training: The following breakdown can be used for the program.

Session: 1 = 30 min	Study unit	Day	Type
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
	Summative assessment	Day___	

Time allocation

Block program: The block program is only a guideline to indicate what will be facilitated . It is important to note that the block program can be changed depending on the pace of the learners and how many learners attended the learning program.

DAY	08:00 TO 08:30	08:40 TO 09:10	09:20 TO 10:00		10:30 TO 11:00	11:10 TO 11:40	11:50 TO 12:30		13:00 TO 13:30	13:40 TO 14:10	14:20 TO 15:00		15:15 TO 16:00
1				T E A				L U N C H				T E A	
2													
3													
4													
5													
6													
7													
8													
9													
10													
11													
12													
13													
14													

Learning event plan

4

Use the format in table beneath to structure the content for a learning event. This provides guidelines for drawing up a facilitation plan in preparation for a presentation. An example of a facilitation plan is indicated in the next table.

Introduction	
Topic	♦ Select a suitable topic. ♦ Introduce the topic verbally. ♦ Display the topic visually.
Interest stimulus	♦ The aim of the interest stimulus is to obtain the learners' interest in the topic under discussion. ♦
Outcomes	♦ The outcome(s) state what the learners should be able to do after the learning event. ♦ Display the outcomes visually. ♦
Scope	♦ The scope of the presentation consists of all the main headings which will be covered in the theory lesson. ♦ Display the scope visually. ♦ Present the scope in a logical sequence. ♦
Importance	♦ The importance of the theory lesson must be stressed so that the learners realize how they will benefit by attending it. ♦
Ground rules	♦ The aim of establishing ground rules is to allow the learners to produce their own set of rules. ♦ Allow the learners to commit themselves to the ground rules by doing an observable activity. ♦
Physical environment	♦ The physical environment must be effective for learning. It includes aspects such as seating arrangement and equipment. ♦

Contents (Body)

Scope point no 1

The first scope point of the presentation will now be discussed in detail. Cover the rest of the scope points one by one.



- Establish a positive relationship with the learners (e.g. professional behavior).
- Indicate throughout the presentation which scope point is being discussed.
- Use at **least two relevant learning aids** (e.g. video, posters).
- The learning aids must support the theory.
- Present the theory lesson on the level of the target group.
- Explain the theory component by making use of appropriate examples.
- Allow the learners to ask questions or to comment.
- Apply effective verbal communication skills (e.g. speak loud enough, vary tone of voice).
- Apply effective non verbal communication skills (e.g. avoid irritating gestures, keep eye contact).
- Apply effective micro-skills (e.g. active listening, paraphrasing, summarizing).
- Control breathing so that voice usage is not affected by shortness of breath.
- Obtain maximum learner participation by including activities and group discussions.
- Keep control over disruptive influences (e.g. learners are not allowed to make unsuitable remarks and comments).
- Allow time for learners to ask questions during the presentation.

Closure

Summary

Questions

Summary

- ♦ Ask questions that are relevant to the outcomes stated.
 - ♦ Use different types of questions (e.g. close-ended, open-ended).
 - ♦ Avoid the use of questions such as leading, subjective, double-barreled etc.
 - ♦ Direct the questions to all the learners, instead of individuals.
 - ♦ Reinforce the questions and answers of the learners.
 - ♦ Provide positive feedback in time.
 - ♦ Allow activities to be performed to enable learners to reflect on the content.
 - ♦ Apply the theory to the practical situation (e.g. giving an example).
-
- ♦ Assess the learners by asking well selected questions concerning all the important aspects.

Assess



(It is very important that these questions are formulated in your planning phase and written into your presentation plan. It is thus determined whether the learners have achieved the outcome(s) of the theory lesson.)

Interest stimulus

- ♦ If there is a forthcoming presentation an interest stimulus must also be given to stimulate the interest of the learners for the next theory lesson.

Learning event plan

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Study unit: _____				
Learning Event: _____				
Topic: _____				
Series No	Facilitator activities	Learner activities	Learning aids	Time
	INTRODUCTION			

Learning event plan

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Series No	Facilitator activities	Learner activities	Learning aids	Time
	Interest stimulus:			
	Outcomes: ♦ ♦ ♦ ♦ ♦			
	Scope:			
	Importance			

Learning event plan

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Series No	Facilitator activities	Learner activities	Learning aids	Time
	Ground Rules			
	Scope point no 1: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 2: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 3: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 4: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 5: (Contents)			

Learning event plan

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Series No	Facilitator activities	Learner activities	Learning aids	Time
	Summary:			

Learning event plan

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Series No	Facilitator activities	Learner activities	Learning aids	Time
	Assess:			
	Interest stimulus:			

Program evaluation questionnaire

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Name of Facilitator: _____

Date: _____

Please give your honest reaction and comments. Mark the appropriate answer with an X

		Disagree	Neutral	Agree
LEARNING PROGRAM				
1.	The learning material is neatly compiled.			
2.	I received all the learning material in time.			
3.	The module content addressed my learning needs.			
4.	The exercises in the module enabled me to understand the theory.			
5.	The self assessments in the module enabled me to reach the stated outcomes.			
6.	The written assignment enabled me to master the content.			
7.	I could apply the theory into my work environment.			
LEARNING ENVIRONMENT				
8.	The venues were adequately prepared.			
9.	Suitable learning aids were used.			
LEARNING FACILITATORS				
10.	The learning facilitators demonstrated courteous behavior during training.			
11.	The learning facilitators were supportive towards me.			
12.	The learning facilitators appeared knowledgeable on the subject.			
13.	The learning facilitators demonstrated effective facilitation skills.			
14.	Enough time was allocated for me to practice facilitation skills.			
15.	I received adequate feedback on the progress that I made.			

Program evaluation questionnaire

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PROGRAM CONTENT										
		1	2	3	4	5	6	7	8	9
16.	How do you rate the subject content?									
17.	How do you find the quality of this program to be?									
18.	Rate the program for the qualities listed:									
	• Practical value									
	• Thoroughness									
	• New ideas gained									
	• Maintain your interest									
	• Helpful to developing yourself									
	• Clear, understandable.									
	• Did you understand the outcomes of the program?									

GENERAL COMMENTS BY LEARNER:

ACTION TAKEN BY FACILITATOR:

LEARNER GUIDE

Unit Standard Title: Investigate life and work related problems using data and probabilities

Unit Standard No: 9012

Unit Standard Credits: 5

NQF Level: 3

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This material has been compiled based on the **UNIT STANDARD Investigate life and work related problems using data and probabilities**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by:

Training Dynamics Africa

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1. Introduction

This learning program is part of a complete qualification. For more details concerning the complete qualification please contact your service provider or alternatively the office of the SASSETA

2. Purpose of this learning program

This Unit Standard is designed to provide credits towards the mathematical literacy requirement of the NQF at Level 3. The essential purposes of the mathematical literacy requirement are that, as the learner progresses with confidence through the levels, the learner will grow in: a confident, insightful use of mathematics in the management of the needs of everyday living to become a self-managing person

An understanding of mathematical applications that provides insight into the learner's present and future occupational experiences and so develop into a contributing worker. The ability to voice a critical sensitivity to the role of mathematics in a democratic society and so become a participating citizen.

People credited with this Unit Standard are able to:

- ♦ Pose questions, collect and organise data.
- ♦ Represent and interpret data using various techniques to investigate real life and work problems.
- ♦ Use random events to explore and apply probability concepts in simple life and work related situations.

2.1 Target group

This program is compiled for the following **target group**:

- ♦ Security members
- ♦ South African Defense Force members
- ♦ South African Police Force members
- ♦ Correctional Services
- ♦ Individuals who wishes to complete the NQF level 3 National Certificate in security practices.

3. Standards and qualifications

It is important that you as the learner understand and realize that this training program is not presented in isolation, but that it was developed based upon nationally recognized standards known as unit standards.

Unit standards are the "building blocks" of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF).

REMEMBER.....

Unit standards comprises of outcomes. An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention.

4. Assessments

In order to assess whether a learner can actually demonstrate the desired outcomes, assessment criteria are included in the unit standard. Each outcome has its own set of assessment criteria.

REMEMBER.....

The assessment criteria describe the evidence that is needed that will show that the learner has demonstrated the outcome correctly.

Kindly refer to the unit standard attached hereto for the assessment criteria.

It is of utmost importance that the learner fully understands the assessment criteria as listed in the unit standard, as it is the only way in which the learner will know what he will be assessed against.

The **final or summative assessment** is the most important aspect of this training program. It is during this process that the learner will be declared competent or not yet competent.

The learner will know exactly how he will be assessed, and when and where he will be assessed. All of these details must be obtained from the training provider where the learner enrolled for his program.

5. Range statements

REMEMBER.....

Also included in the unit standard are the range statements in support of the assessment criteria.

The range statements indicate detailed requirements of the assessment criteria.

6. The learning guide

The learner guide is included in this material under various learning units. The learner guide has been designed in such a manner that the learner is guided in a logical way through the learning material and requirements of the unit standard.

On completion of this skill program, the learner will be assessed against the assessment criteria as stipulated in the unit standard.

Each of the modules consists of learner outcomes to be achieved during training and development interventions, work related experience or at your own time. The learner outcomes will enable you to achieve the required level of confidence and competence in order to undergo the summative (final evaluations) assessment.

The learner needs to assess his own knowledge and skill throughout the training process by completing the learner workbook. With the completion of the workbook you start collecting evidence to proof your competence. The learner workbook will be assessed (FORMATIVE ASSESSMENT)

TAKE NOTE

The learner guide will remain the property of the learner once the LEARNING PROGRAM has been completed.

TAKE NOTE

- The best results will be obtained if you start with Study Unit 1 in the Learner Guide, and work your way through it, study unit by study unit.
- All learning outcomes are vital and must be studied (and exercised for all practical requirements) thoroughly to ensure that enough evidence for all specific outcomes is generated.

7. Learning assumed to be in place

It is assumed that a learner which registers for this program have completed a program in communication at NQF Level 3 or equivalent.

It is imperative to understand that this program are only one of the building blocks to complete Skills program 1 The complete list of unit standards for Skills program appears on the next page.

7.1 Security program matrix
UNIT STANDARDS:

	ID	UNIT STANDARD TITLE	LEVEL	CREDITS
Core	244176	Use security equipment	Level 2	2
Core	114941	Apply knowledge of HIV/AIDS to a specific business sector and a workplace	Level 3	4
Core	244184	Apply legal aspects in a security environment	Level 3	8
Core	244177	Conduct a security patrol in an area of responsibility	Level 3	7
Core	117705	Demonstrate knowledge of the Firearms Control Act 2000 (Act No 60 of 2000) applicable to possessing a firearm	Level 3	3
Core	246694	Explain the requirements for becoming a security service provider	Level 3	4
Core	244182	Give evidence in court	Level 3	4
Core	244179	Handle complaints and problems	Level 3	6
Core	244181	Perform hand over and take over responsibilities	Level 3	2
Core	244189	Conduct access and egress control	Level 4	7
Core	242825	Conduct evacuations and emergency drills	Level 4	4
Core	11505	Identify, handle and defuse security related conflict	Level 4	12
Fundamental	119472	Accommodate audience and context needs in oral/signed communication	Level 3	5
Fundamental	9010	Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations	Level 3	2
Fundamental	9013	Describe, apply, analyse and calculate shape and motion in 2- and 3-dimensional space in different contexts	Level 3	4
Fundamental	119457	Interpret and use information from texts	Level 3	5
Fundamental	9012	Investigate life and work related problems using data and probabilities	Level 3	5
Fundamental	119467	Use language and communication in occupational learning programmes	Level 3	5
Fundamental	7456	Use mathematics to investigate and monitor the financial aspects of personal, business and national issues	Level 3	5
Fundamental	119465	Write/present/sign texts for a range of communicative contexts	Level 3	5
Elective	117889	Demonstrate an understanding of Victim Empowerment	Level 2	6
Elective	12484	Perform basic fire fighting	Level 2	4
Elective	13912	Apply knowledge of self and team in order to develop a plan to enhance team performance	Level 3	5
Elective	244178	Apply neighbourhood watch principles	Level 3	5
Elective	113852	Apply occupational health, safety and environmental principles	Level 3	10
Elective	243188	Care for a service dog	Level 3	15
Elective	116534	Carry out basic first aid treatment in the workplace	Level 3	2

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Elective	113909	Coach a team member in order to enhance individual performance in work environment	Level 3	5
Elective	114942	Describe how to manage reactions arising from a traumatic event	Level 3	2
Elective	243190	Handle a trained service dog to deter crime	Level 3	8
Elective	119649	Handle and use a handgun	Level 3	2
Elective	119651	Handle and use a manually operated rifle or carbine	Level 3	2
Elective	119650	Handle and use a self-loading rifle or carbine	Level 3	2
Elective	119652	Handle and use a shotgun	Level 3	2
Elective	114979	Operate a computer workstation in a business environment	Level 3	2
Elective	244335	Conduct security at an event	Level 4	5
Elective	252174	Demonstrate an understanding of security measures within a port security environment	Level 4	4
Elective	246693	Explain the requirements for safeguarding a national key point	Level 4	4
Elective	123515	Handle and use a handgun for business purposes	Level 4	3
Elective	123519	Handle and use a manually operated rifle or carbine for business purposes	Level 4	3
Elective	123511	Handle and use a self loading rifle or carbine for business	Level 4	3
Elective	123514	Handle and use a shotgun for business purposes	Level 4	3
Elective	11985	Inspect, drive and maintain an official vehicle	Level 4	3
Elective	11513	Operate effectively within a specified control room environment	Level 4	15

8. Preparation of summative assessment

After you have worked through the **LEARNER GUIDE** and you are satisfied that all theoretical and practical requirements/ evidence, as stipulated by the UNIT STANDARD, can be met, you need to make your own necessary arrangements with the assessor and/or facilitator. The training establishment will then arrange a suitable date for you to attend your pre-assessment meeting and then arrange for a suitable date to complete the final assessment.

8.1 RPL assessment



The assessment of RPL learners will be conducted in the same way as for those of new learners. The assessment pack is exactly the same and will therefore be used for new learners as well as RPL Learners. It must however be noted that learners who are applying for RPL must provide proof of previous learning and subject related experience prior to the assessment.

This proof or evidence can be in the format of certified copies (certificates) of previous learning programs that have been attended.

All the evidence will be assessed and authenticated before a learner will be allowed to enroll for an RPL program.

9. Roles and responsibilities

9.1 Facilitator/instructor

- Prepare to facilitate.
- Transfer of knowledge and skills

9.2 The role of the assessor

- Plan for the assessment
- Prepare the learner for the assessment

9.3 The role of the Learner

- Prepare for assessment.
- Accept the responsibility of learning.
- Accept the responsibility to produce/deliver evidence.

STUDY UNIT 1

Data

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Explain the term sampling when producing data.
- ♦ Explain what the term sampling methods.
- ♦ Explain what surveys are and how to use it during the data producing phase.
- ♦ Produce data within your own work environment by using the information in this learner guide.
- ♦ Organise and analyse data that was produced within your work environment.
- ♦ Display data and distributions as set out in this guide.
- ♦ Represent data with respect to centre, spread and distribution.
- ♦ Display data and distributions as set out in this guide.
- ♦ Explain what measuring data with respect to validity, accuracy, and measurement scales is.
- ♦ Draw a conclusion from your analysis.
- ♦ Explain what index numbers are.
- ♦ Explain what simple probability is

1. Definition

DEFINITION

The word '**data**' means a series of observations, measurements or facts given as information. It is a gathered body of facts. The use of data enables you to something else, e.g. predict the weather.

DEFINITION

Probability is the likelihood or chance that something is the case or will happen.

2. Data Collection

There are many methods of collecting primary data and the main methods include:

- ♦ Questionnaires
- ♦ Interviews

DEFINITION

Observation: Observation involves recording the behavioural patterns of people, objects and events in a systematic manner.

DEFINITION

Case-studies: Usually refers to a fairly intensive examination of a single unit such as a person, a small group of people, or a single company and involves measuring what there is and how it got there.

DEFINITION

Diaries: A way of gathering information about the way individuals spend their time on professional activities.

DEFINITION

Critical incidents: Collect reports about what people do that is particularly effective in contributing to good performance.

DEFINITION

Portfolios: Recording information about how each problem arose, methods used to solve it, difficulties encountered, etc.

Let us look at the two most commonly used methods:

2.1 Questionnaires

Questionnaires are a popular means of collecting data, but are difficult to design and often require many rewrites before an acceptable questionnaire is produced.

Advantage:

- ♦ Can cover a large number of people or organisations.

Disadvantage:

- ♦ Questions have to be relatively simple.

2.2 Interviews

Interviewing is a technique that is primarily used to gain an understanding of the underlying reasons and motivations for people's attitudes, preferences or behaviour. Interviews can be undertaken on a personal one-to-one basis or in a group. They can be conducted at work, at home, in the street or in a shopping centre, or some other agreed location.

Personal interview

Advantage:

- ♦ Serious approach by respondent resulting in accurate information.
- ♦ Disadvantage:
- ♦ Time consuming.

Focus group interviews

A focus group is an interview conducted by a trained moderator in a non-structured and natural manner with a small group of respondents. The moderator leads the discussion. The main purpose of focus groups is to gain insights by listening to a group of people from the appropriate target market talk about specific issues of interest.

3. Representing Data

Data may be represented in the form of tables or graphs.

3.1 Table



A **table** is a set of data elements (values) that is organized using a model of vertical columns (which are identified by their name) and horizontal rows. A table has a specified number of columns, but can have any number of rows.

3.2 Graphs

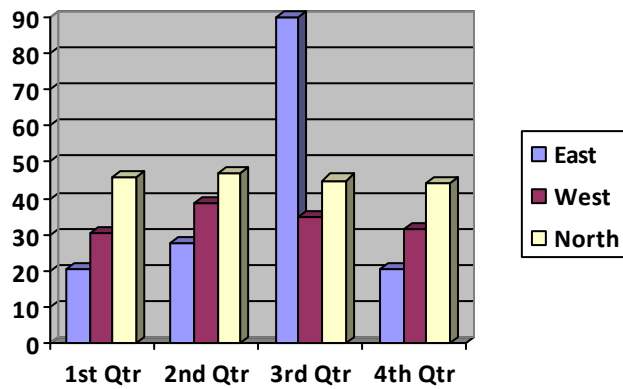


A graph is a set of objects called *points*, *nodes*, or *vertices* connected by links called *lines* or *edges*.

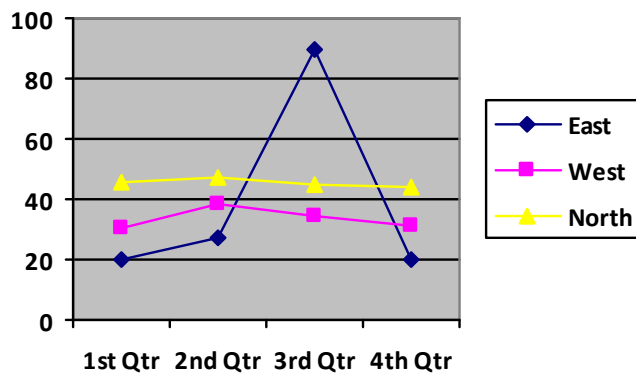
Graphs and charts are great because they communicate information visually. For this reason, graphs are often used in newspapers, magazines and businesses around the world.

The most widely used graphs are bar graphs, line graphs, area graphs, XY graphs and pie graphs/charts.

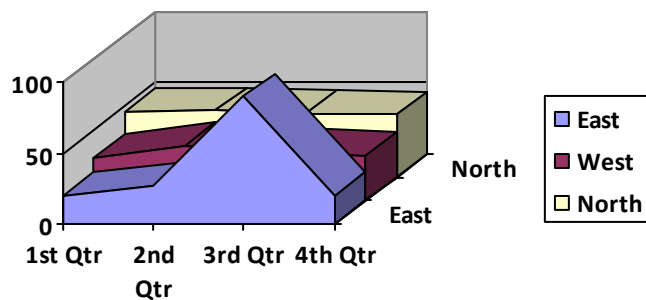
Bar

**EXAMPLE:**

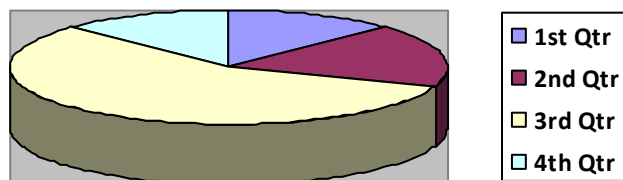
Line

**EXAMPLE:**

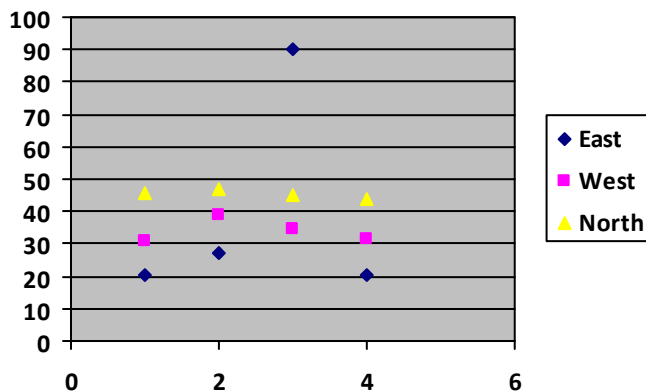
Area

**EXAMPLE:**

Pie

**EXAMPLE:**

XY



EXAMPLE:

TAKE NOTE

All graphs, except the pie graph, are drawn on axes called x (horizontal) and y (vertical). x and y are variables that depend on each other. Normally we treat y as depending on x, e.g. $y = x$ or $y = 2x$. The steepness of the line is called the gradient.

Scatter Graphs

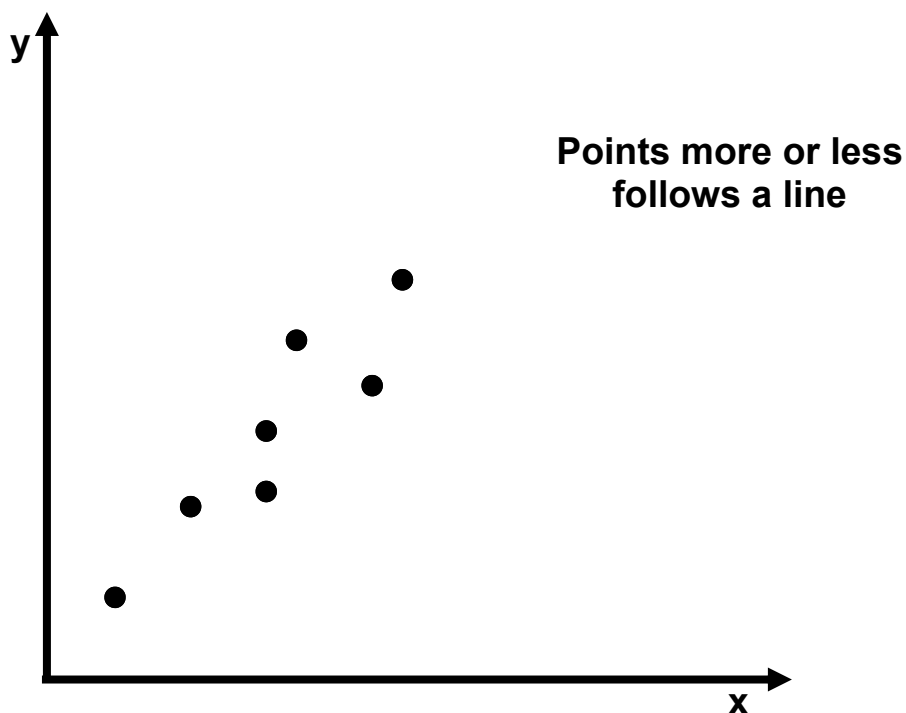
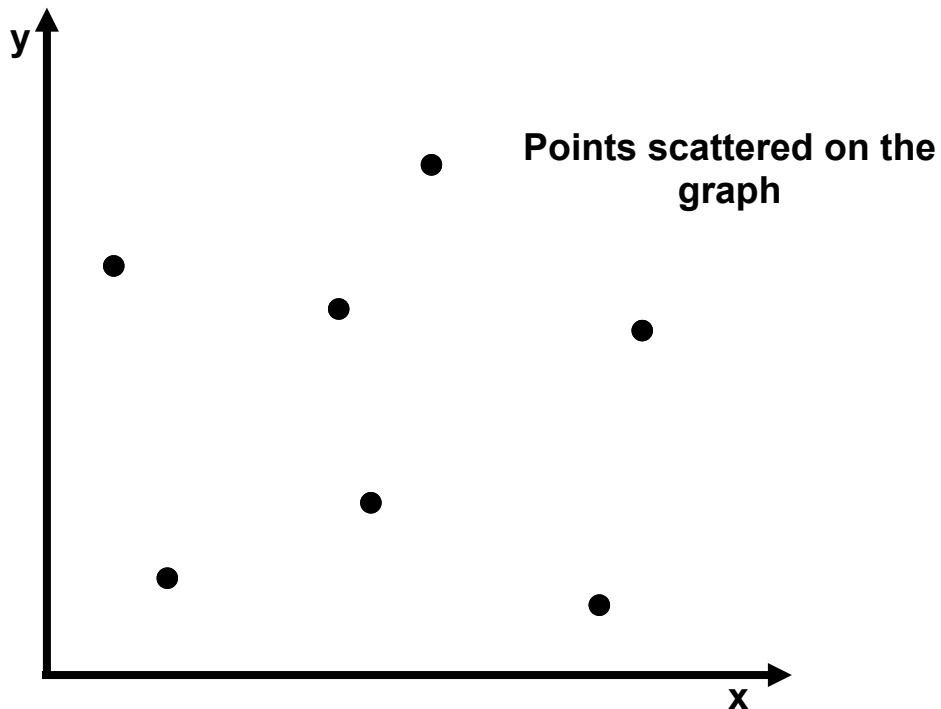
Another very useful tool to represent and analyse data is the scatter graph. This can be used when there is paired information and we want to establish whether a relationship exists between two sets of information (variables).

EXAMPLE:

Examples of variables where a relationship might exist include:

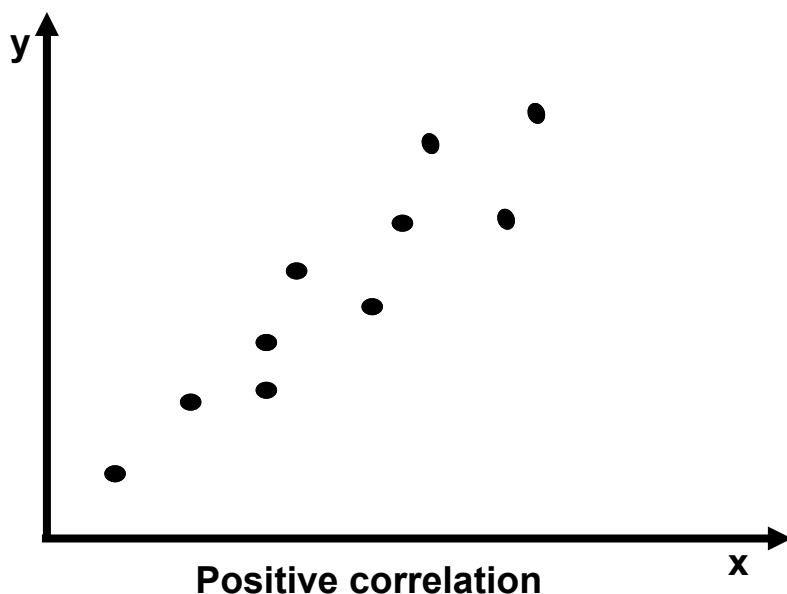
- ♦ The number of sunshine hours and the temperature.
- ♦ Size of a house and monetary value of a house.
- ♦ Amount of fertilizer and the amount of growth of a mealie plant.
- ♦ Height of a person and their shoe size.

Once you have drawn the scatter graph, you can examine the graph to see if a relationship exists between the points. The points might be randomly scattered on the graph or they may more or less follow a line.

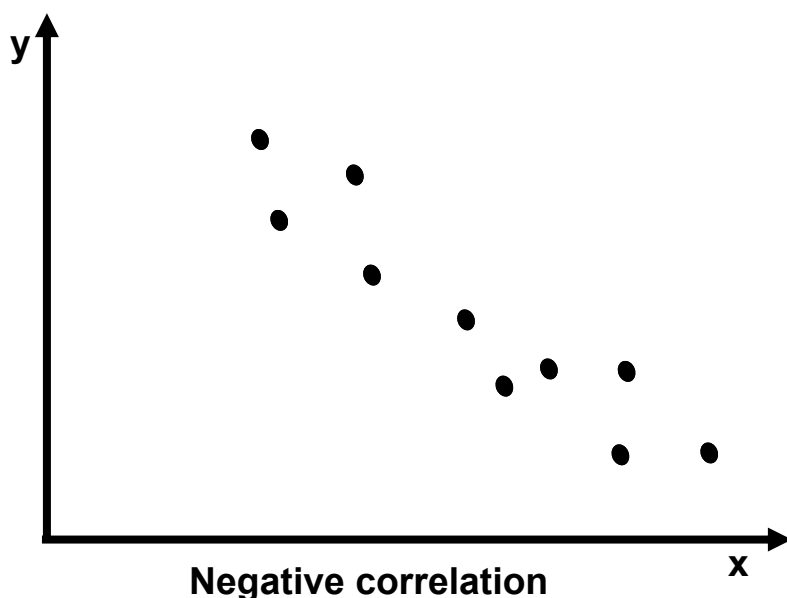


If the points on the scatter graph lie within a narrow band then we say that there is a relationship between the variables. We call this relationship a **correlation**. The correlation may be positive or negative.

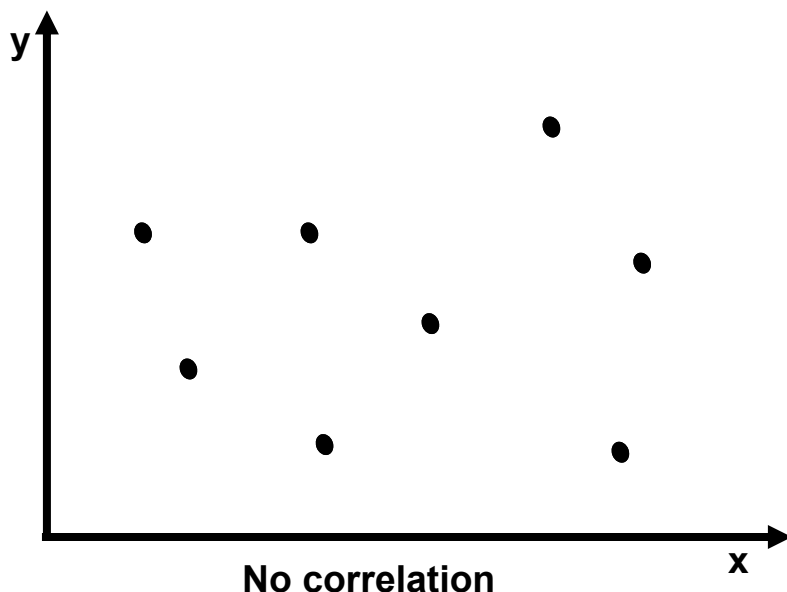
If the points on the scatter graph look something like below, with the band sloping *upwards* across the graph, the points show a **positive correlation**. This means that as the one variable increases, the other variable also increases.



If the points on the scatter graph look something like below, with the band sloping *downwards* across the graph, the points show a **negative correlation**. This means that as the one variable increases, the other variable decreases.



If the points on the scatter graph look something like below, with the points *scattered all over*, we say that there is **no correlation** between the two variables. The one variable is not connected to the other at all.



Using the above scatter graph we would be unable to make any predictions about the variables.

If the variables are related, then any change in the value of one of the variables will be associated with a change in value of the other variable.

EXAMPLE:

For example, there is a correlation between the number of kilometers you travel and the amount of petrol your car will consume, because the further you travel, the more petrol your car will consume.

It is useful to know whether there is a correlation between variables, because such knowledge allows you to make plans, predictions and decisions concerning the variables.

If we took a sample of people and recorded their height and shoe size, we would expect a graph of height against shoe size to look something like this:

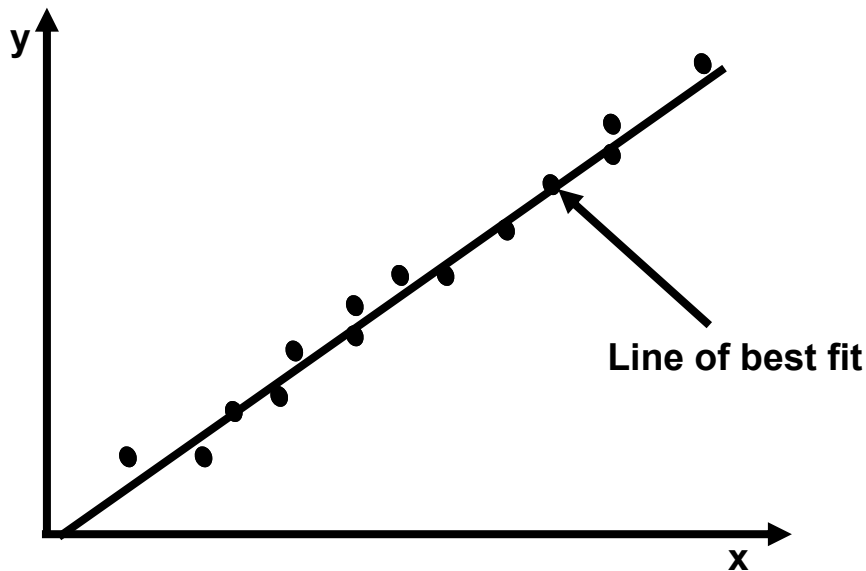


From the scatter graph we can see that there **is** a relationship between height and shoe size. As a person's height increases, the size of their shoes also increases. This is called a positive correlation and we could use this correlation, and a **line of best fit**, to *make predictions* about the shoe size a person measuring a certain height would wear.

3.3 Line of best fit

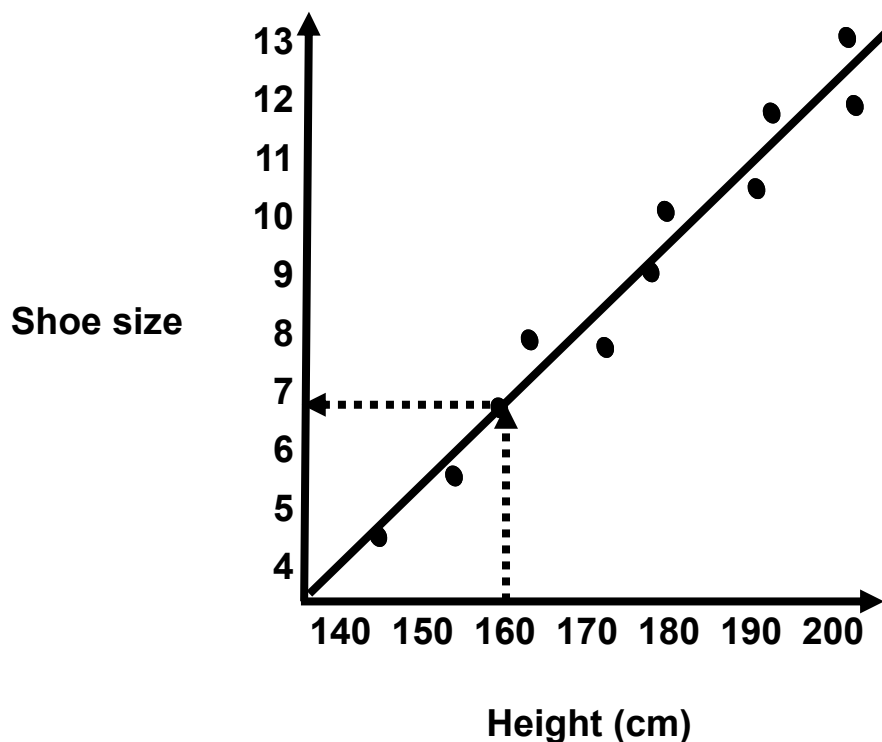
To show correlation between points more clearly, we can draw a line of best fit. The line of best fit shows the general trend of the points on the graph. You can use a transparent plastic ruler to judge where to put the line. Place the ruler across the points. Move it until one of its edges fits the points best. Then you can draw the line. Try to make the line:

- ♦ match the slope of the plotted points
- ♦ have as many plotted points on the line as possible
- ♦ have the same number of plotted points above the line as below the line



The line of best fit is an approximation or estimate. The slope of the line of best fit might vary depending on the person drawing it. You can use the line of best fit to predict values.

Relationship between height and shoe size



Using the line of best fit, we can predict the size shoe of a boy with a height of 160 cm, by looking for the corresponding shoe size. From the graph, this would be a shoe size of approximately 7.

4. Analysing Data

DEFINITION

To analyse means to examine something in detail in order to discover meaning, or to break down into components or essential features. In order to achieve this in Maths, we often have to *compare* data.

When we have to compare two lists of data, there are two types of measurements that are useful to know. One is the **Measure of Central Tendency** (Mean, Median and Mode) and the other is the **Measure of Dispersion** (Range, Interquartile Range and Standard Deviation).

4.1 Measures of Central Tendency

There are three measures of central tendency: the mean, the median and the mode.

- ♦ **The MEAN** is found by adding together all the data items and then dividing by the number of data items.
- ♦ **The MEDIAN** is the middle data item after all the data items have been listed in ascending order. If there are an even number of data items, the median is the *average* of the 2 middle data items.
- ♦ **The MODE** is the most common data item after all the data items have been listed in ascending order.

EXAMPLE:

The heights, in centimeters, of 12 learners, written in ascending order are:

126 126 128 129 131 140 145 145 145 157 169 210

Find:

- ♦ the mean height
- ♦ the median height
- ♦ the mode height

Solution:

- ♦ **Mean** =
$$\frac{126+126+128+129+131+140+145+145+145+157+169+210}{12}$$
$$= \frac{1\,751}{12}$$
$$= 145,916... \text{ cm} \approx 145,9 \text{ cm}$$

- ♦ As there are 12 data items, the median will lie between the 6th measurement (140 cm) and the 7th measurement (145 cm).

$$\begin{aligned}\text{So the median} &= \frac{140 + 145}{2} \\ &= 142,5 \text{ cm}\end{aligned}$$

- ♦ There are two heights of 126 cm and THREE heights of 145 cm.

$$\text{So the mode} = 145 \text{ cm}$$

4.2 Measures of Dispersion

These tell us about the way the data is dispersed or spread.

- ♦ The simplest measure of dispersion is the RANGE.



The range = largest value – smallest value.

The range is not often used, however, as it can be affected by values that are either very large or very small and not representative of the rest of the values in a data set.

- ♦ The measure of dispersion around the median is called the INTERQUARTILE RANGE.



The interquartile range (IQR) = the upper quartile (Q₃) – the lower quartile (Q₁). The IQR is the range of the middle 50% of the data items.

The measure of dispersion around the mean is called the STANDARD DEVIATION. We can find the standard deviation using the stats mode on a scientific calculator, or by using the formula:

$$\sqrt{\frac{\text{standard deviation} = \frac{\text{sum of (deviations from the mean)}^2}{\text{number of data items}}}$$

EXAMPLE:

The heights, in centimeters, of 12 learners, written in ascending order are:

126 126 128 129 131 140 145 145 145 157 169 210

Find:

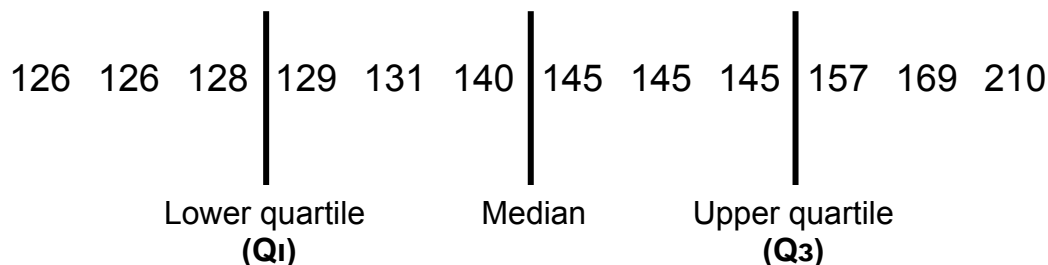
- ♦ the range
- ♦ the interquartile range
- ♦ the standard deviation

Solution:

- ♦ **Range** = largest value – smallest value = 210 cm – 126 cm
= 84 cm

- ♦ In order to find the interquartile range, we find the median, the lower quartile (Q_1) and the upper quartile (Q_3).

A quartile divides a set of data into four equally sized sets of data. From the previous example we know that the median (M) = 142,5 cm. The median lies between the 6th and 7th measurement.



⇒ The lower quartile lies between the 3rd and 4th measurement.

⇒ The lower quartile (Q_1) = $\frac{128 + 129}{2} = \frac{257}{2} = 128,5$ cm

⇒ The upper quartile lies between the 9th and 10th measurement.

⇒ The upper quartile (Q_3) = $\frac{145 + 157}{2} = \frac{302}{2} = 151$ cm

So the interquartile range (IQR) = (Q_3) – (Q_1)
= 151 cm – 128,5 cm
= 22,5 cm

EXAMPLE:

- ♦ Use a calculator to find the standard deviation. The following is the key sequence for the Casio fx82-ES:

[MODE] [2:STAT] [1:1-VAR] 126 [=] 126 [=] 128 [=] 129 [=] 131 [=] 140 [=] 145 [=] 145 [=] 145 [=] 157 [=] 169 [=] 210 [=] [AC] [SHIFT] [STAT] [5:VAR] [3: σn] [=]

You should find that the standard deviation = 23,135...cm $\approx$ 23,1 cm

5. Probability

DEFINITION

As said previously, probability is the likelihood or chance that something is the case or will happen.

To determine probability, we use this formula:

$$P(A) = \frac{\text{The total number of ways a specific event (A) can occur}}{\text{The total number of possible outcomes}}$$

P = probability

A = the event

The probability of event A happening is the number of ways event A can occur divided by the total number of possible outcomes.

EXAMPLE:

There are 2 possible outcomes when a coin is flipped: heads or tails.

- ♦ The probability of getting heads is 1 out of 2 outcomes or $\frac{1}{2}$.
- ♦ We can also write probability or chance as a percentage: There is a 50% chance of getting heads.

EXAMPLE:

Let's calculate the probability of throwing a 5 on a dice. With one throw of one dice, how many ways can you get a 5? **One** (only one 5 on the dice)

- ♦ How many possible outcomes are there? Six (you could get 1,2,3,4,5 or 6)
- ♦ So, the probability of throwing a 5 = $\frac{1}{6}$

EXAMPLE:

- ♦ What is the probability of throwing an odd number on a dice?
- ♦ With one throw of one dice, how many ways can you get an odd number? Three (1, 3 or 5)
- ♦ How many possible outcomes are there? Six (1, 2, 3, 4, 5 or 6)

The probability of throwing an odd number = $\frac{3}{6}$ or $\frac{1}{2}$

We have now looked at very basic/elementary examples of probability. When it comes to more complex/complicated problems, there are various 'tools' available to help us work it out. The most common of these tools is a tree diagram.

DEFINITION

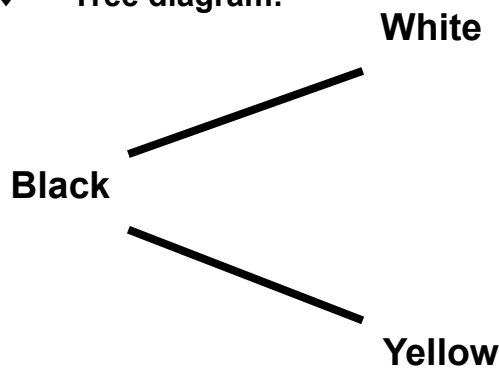
A tree diagram is a branching diagram showing the probability of various events.

EXAMPLE:

- A company is choosing kit for their soccer team. They decide on black shorts but two different shirts, white and yellow. We can show the different combinations by drawing a tree diagram as follows:

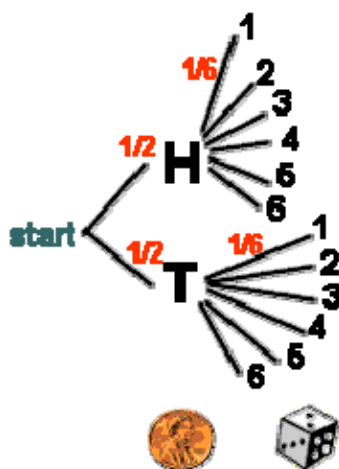
Possible outfit combinations	
Shorts	Shirt
Black	White
Black	Yellow

- Tree diagram:



This gives **two possible combinations**. So the probability of wearing black shorts and a white shirt = $\frac{1}{2}$.

- Show the tree diagram for tossing one penny and rolling one die.
(H = heads, T = tails)



By following the different paths in the tree diagram, we can arrive at the possible outcomes:

H1, H2, H3, H4, H5, H6 and T1, T2, T3, T4, T5, T6

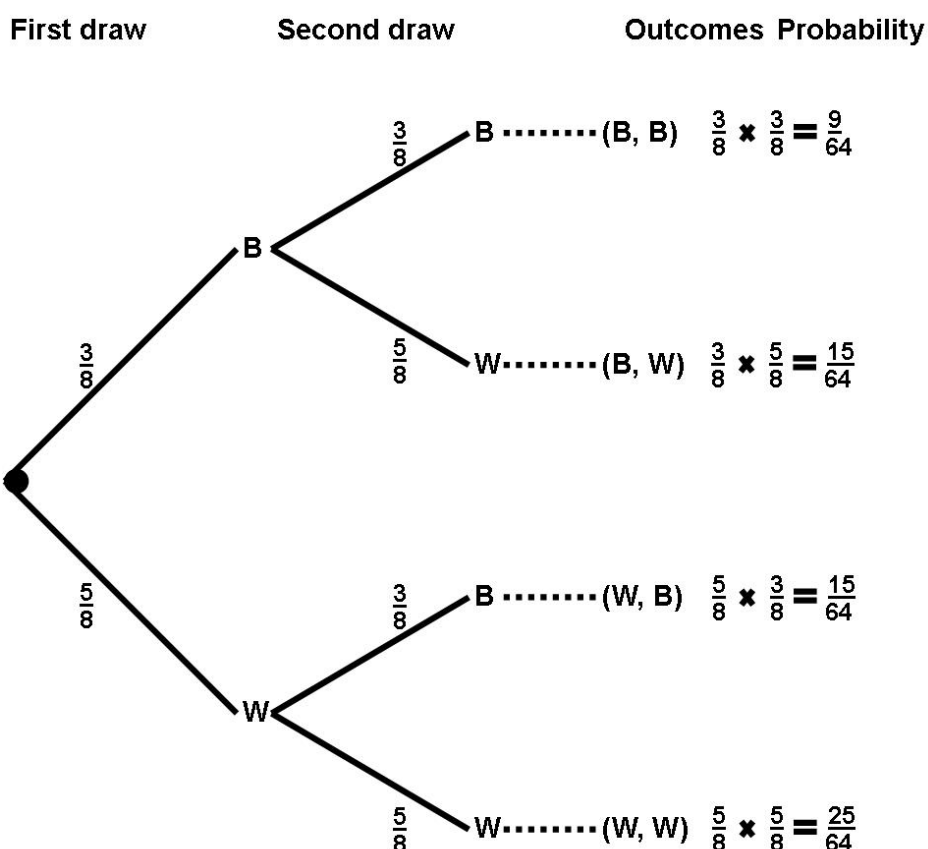
- The probability of each of these outcomes is $\frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$

A **probability tree diagram** shows all the possible events. The first event is represented by a dot. From the dot, branches are drawn to represent all possible outcomes of the event. The probability of each outcome is written on its branch.



- ♦ A bag contains 3 black balls and 5 white balls. Paul picks a ball at random from the bag and replaces it back in the bag. He mixes the balls in the bag and then picks another ball at random from the bag.
- Construct a probability tree of the problem.
- Calculate the probability that Paul picks:
 - ⇒ two black balls
 - ⇒ a black ball in his second draw

Solution:



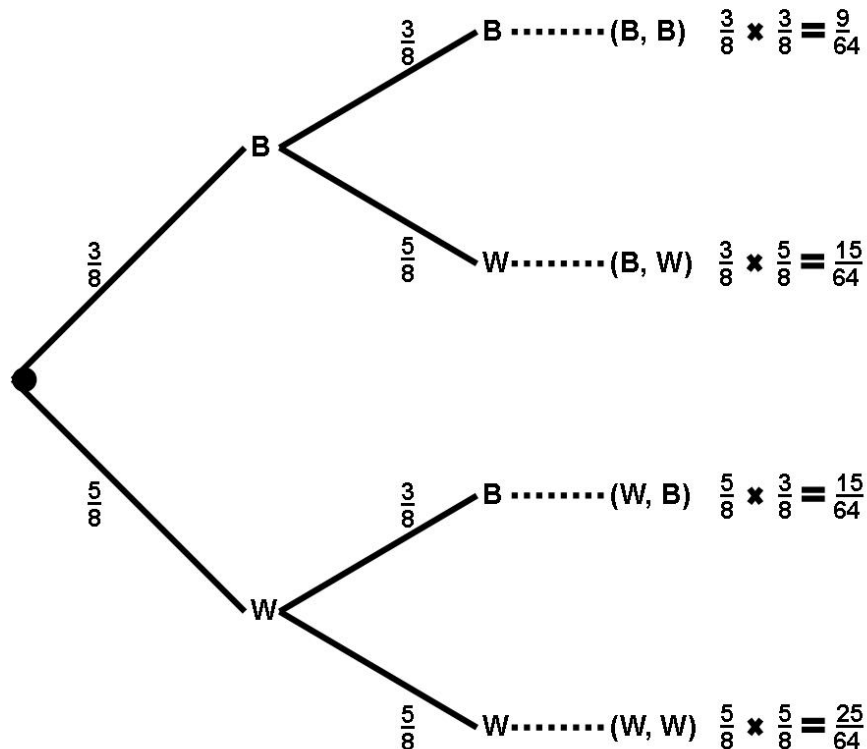
EXAMPLE:

- Check that the probabilities in the last column add up to 1.

$$\frac{9}{64} + \frac{15}{64} + \frac{15}{64} + \frac{25}{64} = \frac{64}{64} = 1$$

- To find the probability of getting two black balls, first locate the B branch and then follow the second B branch. Since these are independent events we can multiply the probability of each branch.

First draw Second draw Outcomes Probability

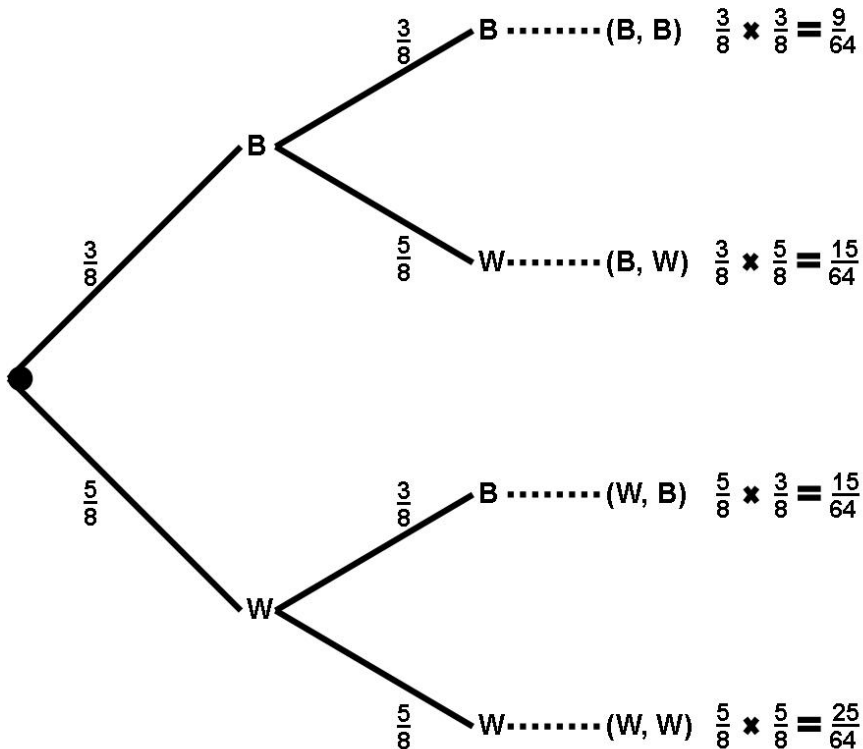


$$P(\text{two black balls}) = \frac{3}{8} \times \frac{3}{8} = \frac{9}{64}$$



- ♦ There are two outcomes where the second ball can be black. Either (B, B) or (W, B)

First draw Second draw Outcomes Probability



From the probability tree diagram, we get:

$$P(B, B) = \frac{9}{64} \text{ and } P(W, B) = \frac{15}{64}$$

$$P(\text{second ball black}) = \frac{9}{64} + \frac{15}{64}$$

$$= P(B, B) \text{ or } P(W, B)$$

$$= P(B, B) + P(W, B) = \frac{24}{64}$$

$$= \frac{3}{8}$$

[illegible]

WORDS/ACRONYMS	EXPLANATION OF ACRONYMS
ASSESSMENT	“A structured process of gathering evidence and making judgments about an individual’s performance in relation to registered national standards and qualifications”
ASSESSMENT CRITERIA	It describe the evidence that is needed to show that the learner has demonstrated the outcome correctly. A list of evidence to be supplied by the candidate as proof of his / her ability to integrate all of the competencies listed earlier in the standard.
COMPETENCE	Competence is the ability to do certain things in a range of different environments with knowledge and understanding about what is being done and the effect this has on others. It includes <u>Knowledge, skills, attitudes, values and ethics.</u>
CREDIT	Each unit standard is worth a number of credits which gives an indication of the size and scope of the unit standard, the standard is worth. (1 credit = 10 notional hours of learning)
CRITICAL CROSSFIELD OUTCOMES	critical cross field outcomes is to direct educational activities towards development of the learners within a social and economic environment.
EMBEDDED KNOWLEDGE	Body of theory essential for understanding and applying the standard. Principles, methods and techniques that would underpin good practice of this unit standard.
EVIDENCE	Evidence is a set of information or observations that provides proof of the learner’s competence.
FORMATIVE ASSESSMENT	Aims to assist the learner to identify any knowledge/skills gaps when demonstrating his/her current competence. Diagnostic assessment supports learning and progress of the learner.
INTEGRATED ASSESSMENT	Integrated assessment refers to: • assessing a number of outcomes together • assessing a number of assessment criteria together • assessing a number of unit standards together • using a combination of assessment methods and instruments for an outcome/s • collecting naturally occurring evidence in the workplace • Acquiring evidence from other sources such as supervisor’s reports, testimonials, portfolio of work previously done, logbooks, journals etc.
LEARNING ASSUMED TO BE IN PLACE	Building blocks (knowledge, skill and understanding) assumed to be in place
LEARNING PROGRAM	A “learning program is a set of learning activities in which the learner will be involved in working towards the achievement of one or more outcomes.
LEVEL	Each unit standard is registered on the NQF at a particular level based on the complexity of the outcomes and stringency of the standards.
NQF	National Qualifications Framework
RANGE STATEMENTS	The range statements indicate detailed requirements of the assessment criteria.

RE-ASSESSMENT	• The same method and instrument may be used, but, the task and materials should be different • The task and materials should, however, be of the same complexity and level as the previous ones. • In case the methods and instruments are changed it must be ensured that they are appropriate for the outcomes specified.
RPL ASSESSMENT	RPL is a way of recognizing what individuals already know and can do. The basic premise underlying it is that people, especially mature adults, learn many things inside and outside the formal structures of education and training and that irrespective of where, how and when learning was acquired, subject to assessment, it can be worthy of recognition and credit.
RPL	Recognition of prior learning
SASSETA	Safety & security sector education and training authority.
SAQA	SOUTH AFRICAN QUALIFICATION AUTHORITY
SPECIFIC OUTCOMES	An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention. Key performance areas in the standard are given, the focus is <i>not</i> only on current competence, but also on future competence in this standard.
SUMMATIVE ASSESSMENT	Assessment used to record the level of competence achieved by a learner over a period of time, e.g. at the end of a program, term or year.
QUALIFICATIONS	Represent a planned combination of learning outcomes which has a defined purpose and intended to provide learners with applied competence and a basis for further learning
UNIT STANDARDS	the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF). “A statement of the outcomes that the learner must achieve to earn credit. Standards are nationally agreed and internationally comparable.”

Investigate life and work related problems using data and probabilities

SAQA US ID	UNIT STANDARD TITLE		
9012	Investigate life and work related problems using data and probabilities		
ORIGINATOR		REGISTERING PROVIDER	
SGB Math Literacy, Math, Math Sciences L 2 -4			
FIELD		SUBFIELD	
Field 10 - Physical, Mathematical, Computer and Life Sciences		Mathematical Sciences	
ABET BAND	UNIT STANDARD TYPE	NQF LEVEL	CREDITS
Undefined	Regular-Fundamental	Level 3	5
REGISTRATION STATUS	REGISTRATION START DATE	REGISTRATION END DATE	SAQA DECISION NUMBER
Reregistered	2008-01-22	2011-01-22	SAQA 0160/05
LAST DATE FOR ENROLMENT		LAST DATE FOR ACHIEVEMENT	
2012-01-22		2015-01-22	

This unit standard does not replace any other unit standard and is not replaced by any other unit standard.

PURPOSE OF THE UNIT STANDARD

This Unit Standard is designed to provide credits towards the mathematical literacy requirement of the NQF at Level 3. The essential purposes of the mathematical literacy requirement are that, as the learner progresses with confidence through the levels, the learner will grow in:
a confident, insightful use of mathematics in the management of the needs of everyday living to become a self-managing person

An understanding of mathematical applications that provides insight into the learner's present and future occupational experiences and so develop into a contributing worker

The ability to voice a critical sensitivity to the role of mathematics in a democratic society and so become a participating citizen

People credited with this Unit Standard are able to:

Pose questions, collect and organise data.

Represent and interpret data using various techniques to investigate real life and work problems.

Use random events to explore and apply probability concepts in simple life and work related situations.

LEARNING ASSUMED TO BE IN PLACE AND RECOGNITION OF PRIOR LEARNING

The credit value is based on the assumption that people starting to learn towards this unit standard are competent in Mathematics and Communications at NQF level 2.

UNIT STANDARD RANGE

This unit standard includes the requirement to:
Identify issues suited to resolution by statistical methods

Select a suitable sample

Collect and generate data through the use of questionnaires and suitable experiments

Calculate statistics and probability values through the use of calculators

Represent data in the form of tables, charts and graphs

Use statistics and representations of data to argue a resolution of an issue

Interpret statistics, the use of probabilities, and representations of data

Determine probability values

Work with probability in practical situations

Use available technology (i.e. whatever is available for working with data e. g. pencil and ruler, including spreadsheets, graphical calculators) to fit appropriate curves (e.g., linear, quadratic) to data

More detailed range statements are provided for specific outcomes and assessment criteria as needed.

Specific Outcomes and Assessment Criteria:**SPECIFIC OUTCOME 1**

Pose questions, collect and organise data.

OUTCOME NOTES

Specific purposes include:

Determining trends in societal issues such as crime and health.

Identifying relevant characteristics of target groups such as age, range, gender, socio-economic group, cultural belief and performance.

Predicting the likelihood of the occurrence of events.

Considering the attitudes or opinions of people on issues.

OUTCOME RANGE

Techniques include:

The selection of a sample from a population with due sensitivity to issues relating to bias.

The formulation and use of questionnaires and interviews to obtain data for specific purposes related to surveys and censuses.

Use of databases to obtain information (e. g., StatsSA for national census data) and data suited to the resolution of particular issues.

Work with deferent types of measuring instruments and scales such as yes/no (dichotomous) 5 point (Likert), discrete, and continuous variables (e g., temperature).

Evaluation of data gathering techniques and of data collected so that faults and inconsistencies are identified. (e.g., in cases where a person may be counted more than once such as when collecting ST13 data).

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

1. Situations or issues that can be dealt with through statistical methods are identified correctly.

ASSESSMENT CRITERION 2

2. Variables contributing to a problem situation are identified and addressed in data gathering, e.g. crime is related to time of day and location.

ASSESSMENT CRITERION 3

3. Appropriate and efficient methods are used to collect, record and organise data.

ASSESSMENT CRITERION 4

4. Data samples are of adequate size and are representative of the population.

SPECIFIC OUTCOME 2

Represent, analyse and interpret data using various techniques.

OUTCOME NOTES

Represent, analyse and interpret data using various techniques to investigate real-life and work problems

OUTCOME RANGE

Calculation of measures of centre and spread such as mean, median, mode, range and inter-quartile range.

Use of scatter plots and intuitively placed lines of best fit to represent the association between two variables. (Regression analysis not included,).

Fit curves (e.g., linear and quadratic cases) to predict trends.

Use of a variety of representations applicable to the issue being investigated.

Specific purposes include:

Determining trends societal issues such as crime and health;

Identifying relevant characteristics of target groups such as age, range, gender, socio-economic group, cultural belief and performance;

Considering the attitudes or opinions of people on issues.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

1. Graphical representations and numerical summaries are consistent with the data, are clear and appropriate to the situation and target audience.

ASSESSMENT CRITERION 2

2. Different representations of aspects of the data are compared to take a position on the issue.

ASSESSMENT CRITERION 3

3. Calculations and the use of statistics are correct and appropriate to the problem.

ASSESSMENT CRITERION 4

4. Interpretations of statistics are justified and applied to answer questions about the problem.

ASSESSMENT CRITERION 5

5. New questions that arise from the modelling of the data are discussed.

SPECIFIC OUTCOME 3

Use random events to explore and apply, probability concepts in simple life.

OUTCOME NOTES

Use random events to explore and apply, probability concepts in simple life and work related situations

[Note: straightforward applications relevant to the life or work related experiences of the learners should be chosen]

OUTCOME RANGE

Distinguish outcomes, which are equally likely (e.g. spinning a coin, rolling a die) from those that are not (e.g. dropping a drawing pin, spinning a biased coin).

Distinguish between a trial (e.g. a turn at rolling a die), outcome (getting a 6 when the die is rolled) and event (getting any even number when rolling a die -a collection of outcomes).

Interpret probability values expressed as fractions between 0 and 1 or as percentages.

Use the term "odds on" in relation to a probability value (e.g. the odds on getting a 4 when rolling a die are 1 to 5 while the probability of getting a 4 is one sixth).

Distinguish between theoretical (e. g., for a fair coin on the basis of equal likelihood) and experimental probabilities (e.g. for getting a pin to land with its point up or its point down when dropped on the basis of relative frequency after a large number of trials).

Use tree diagrams in representing and working with events.

Use basic counting techniques to determine the number of ways an event can occur. (The formal use of permutations and combinations not expected.)

Distinguish between situations in which probabilities need to be multiplied from those in which probabilities need to be added (e. g., drawing the ace of hearts and the ace of spades as opposed to drawing one or the other).

Make and test predictions about probability in the context of games, real-life situations and the workplace.

ASSESSMENT CRITERIA

ASSESSMENT CRITERION 1

1. Data are gathered, organised, sorted and classified in a suitable manner for further processing and analysis.

ASSESSMENT CRITERION 2

2. Experiments and simulations are chosen appropriately in terms of the situation to be investigated.

ASSESSMENT CRITERION 3

3. Probabilities are determined correctly.

ASSESSMENT CRITERION 4

4. Distinctions are correctly made between theoretical and experimental probabilities.

ASSESSMENT CRITERION 5

5. Predictions are based on validated experimental or theoretical probabilities.

ASSESSMENT CRITERION 6

6. The outcomes of experiments and simulations are communicated clearly.

UNIT STANDARD ACCREDITATION AND MODERATION OPTIONS

Providers of learning towards this unit standard will need to meet the accreditation requirements of the GENFETQA.

Moderation Option:

The moderation requirements of the GENFETQA must be met in order to award credit to learners for this unit standard.

UNIT STANDARD ESSENTIAL EMBEDDED KNOWLEDGE

The following essential embedded knowledge will be assessed through assessment of the specific outcomes in terms of the stipulated assessment criteria. Candidates are unlikely to achieve all the specific outcomes, to the standards described in the assessment criteria, without knowledge of the listed embedded knowledge. This means that the possession or lack of the knowledge can be inferred directly from the quality of the candidate's performance against the standards.

Methods for collecting, organising data and calculating statistics

The meaning of concepts such as centre and spread

Techniques for statistically modelling a situation

Random events, equal likelihood, probability

Critical Cross-field Outcomes (CCFO):**UNIT STANDARD CCFO IDENTIFYING**

Identify and solve problems using critical and creative thinking:

Solve a variety of problems or take a position on issues related to the learner based on data, statistics and probability.

UNIT STANDARD CCFO COLLECTING

Collect, analyse, organise and critically evaluate information:

Collect, organise, and interpret data and statistics to make sense of adult situations.

UNIT STANDARD CCFO COMMUNICATING

Communicate effectively:

Use everyday language and mathematical language to represent data, statistics and probability and to communicate conclusions.

UNIT STANDARD CCFO CONTRIBUTING

Use mathematics:

Use mathematics to, describe and represent and interpret life or work related situations and to solve problems relevant to the learner.